

IVAN MAKARENKO

Product Management | Technical Product & Program Management | Robotics & Automation

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Robotics engineer and M.S. Technology Management candidate who turns customer needs into technical requirements and drives cross functional execution across software, hardware, suppliers, and factory validation, from discovery and requirements through prototype, patent filing, and production.

EXPERIENCE

SourceOne Ltd

Product Manager Intern (Part time Remote to Full time On site)

Hong Kong
Feb to Aug 2026

- Evaluated about **10 automated factories across China**, assessing production capability, process controls, manufacturability, and supplier risk with engineering, quality, and sourcing leaders.
- Coordinated **product validation and supplier reviews** for Mainspring Energy, Toyota USA, and Rough Country, translating customer requirements and factory findings into documented technical issues, risks, and next steps.
- Inspected casting, forging, machining, and laser cutting operations against engineering drawings and quality documentation, converting findings into recommendations for product, engineering, and sourcing teams.

Independent Consulting

Founder & Technology Consultant

Remote
Jun 2019 to Dec 2025

- Built a referral practice serving **20+ clients**, generating \$20K to \$70K in annual revenue with recurring retainers.
- Led **discovery, requirements definition, prioritization, delivery, and iteration** for 10+ software, web, and analytics products across construction, legal services, and retail.

Stantec

Automation Engineering Intern

Chandler, AZ
Jun to Aug 2024

- Reviewed about 90 legacy electrical schematics across 12 projects, identified discrepancies in roughly 25%, and updated the corresponding Revit models ahead of multidisciplinary design reviews.
- Standardized Revit templates and built an Excel tracker that centralized documentation across electrical, mechanical, and civil teams, improving consistency and project visibility.

Additive Desert Designs

New Product Development Intern

Mesa, AZ
Jun to Aug 2023

- Compiled a vehicle trim and fitment database covering 50+ truck models, reducing the research needed to select scan vehicles and confirm part compatibility and shortening design cycles by an estimated 30%.
- Developed truck components from laser scanning and SolidWorks CAD through assembly and testing, then supported layout, fixturing, and controller setup for **two OTC DAIHEN robotic welding cells**, both now in production.

SELECTED PROJECTS

FoldRide Pro

Inventor & Product Lead

New York, NY
2025 to Present

- Designed a foldable electric skateboard around a novel folding architecture and trucks that release without tools, translating **user needs into product requirements**, system architecture, drawings, and STEP files for production.
- Filed a **U.S. provisional patent application** on the folding architecture and truck release system, then partnered with a factory on prototypes, dual motor drive, safety systems, and a retail price under \$999.

Robotic Explorer for Hypothesized Surfaces

NASA Psyche Student Collaboration | ASU Senior Capstone

Mesa, AZ
Jan to Dec 2024

- **Proposed the rover concept the team built**, then owned the full electrical subsystem covering power distribution, wiring, motor control, and integration while supporting mechanical design across a team of four.
- Enabled 360° turns without reversing through independent wheel speed and direction control, placing about 70% of mass in the wheels and replacing hydraulic actuation with vacuum rated springs after NASA Psyche mentor feedback.

EDUCATION

Columbia University, M.S. in Technology Management, School of Professional Studies

Expected Dec 2026

GPA 3.8 | Coursework in Product Management, AI Management, and Digital Transformation

Arizona State University, B.S.E. in Engineering (Robotics), Ira A. Fulton Schools of Engineering

Dec 2024

SKILLS

Product & Program Customer discovery | Requirements & PRDs | Prioritization | Roadmaps | Cross functional execution | Risk tracking | Product validation | Agile & Scrum

Engineering & Manufacturing DFM | NPI | Supplier evaluation & quality | Root cause analysis | 8D | FMEA | BOM & costing | Robotic automation | Mechatronics | Embedded systems | PCB design | Product testing

Tools Python | C/C++ | SQL | MATLAB | SolidWorks | Revit | AutoCAD | Cadence | Jira | Git | Excel

Languages English (Fluent) | Russian (Fluent) | Ukrainian (Fluent)